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REVIEW

Prevalence and variability of restrictive care practice use (physical restraint, seclusion and chemical restraint) in adult mental health inpatient settings: A systematic review and meta-analysis

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Abstract

Background: There is a growing consensus to reduce the use of restrictive care practices in mental health settings to minimise the physical and psychological complications for patients. However, data regarding restrictive care practice use and factors contributing to variations in the proportion estimates has not previously been synthesised.

Aims: This study aimed to synthesise evidence on (1) the pooled proportions of physical restraint, seclusion or chemical restraint in adult mental health inpatients and (2) sources of variability in these proportion estimates.

Methods: Studies were identified from Scopus, MEDLINE, PsycINFO, Web of Science, Embase and CINAHL databases following the PRISMA 2020 guidelines. We conducted a meta-analysis of studies published in English language from 1 January 2010 to 15 August 2022. Binomial data were pooled using a random effect model, with 95% confidence intervals. Meta-regression was also computed to identify factors that may contribute to variations in the proportion estimates.

Results: A total of 77 studies were included in this meta-analysis. The pooled prevalence of physical restraint, seclusion and chemical restraint was 14.4%, 15.8% and 25.7%, respectively. Data were heterogeneous across studies ($I^2 > 99\%$). Reporting practices and geographical locations contributed to the variability in the reported estimates of restrictive care practices, with studies from Asian countries reporting higher proportions.

Conclusion: There appear differences between geographical locations in the proportion of restrictive practices in mental health inpatients; however, this is complicated by how these prevalence data have been measured and defined. Consistency in the

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reporting of restrictive care practices in mental health is required to make valid comparisons between geographical regions, policy settings and practice innovations.

Relevance to Clinical Practice: Efforts are needed to develop training programmes and policy changes to ensure consistency in defining and reporting of restrictive care practices in mental health facilities.

Patient/Public Contribution: This is a systematic review that analysed data from previously published studies, and there was no patient/public contribution in this study.

Protocol Registration: The protocol for this review has been registered to PROSPERO: CRD42022335167.

KEYWORDS
coercion, inconsistency, mental health, psychiatry, restraint, restrictive care, seclusion, sedation, variability

1 | INTRODUCTION

In mental health facilities, restrictive care practices (RCPs) have been regarded either as therapeutic interventions to provide psychological and informational support to patients or to reduce violence or risk of harm to the patient, other people or the physical environment since the ancient history of mental health care (Chien et al., 2005; Kaltiala-Heino et al., 2003; Masters, 2017). Restrictive care practices are physical/mechanical, or pharmacological interventions that reduce/limit a person's freedom of movement, ability to make decisions, autonomy or choice (Funk & Bold, 2020; McSherry & Maker, 2020).

Some forms of RCP include seclusion (involuntary confinement of an individual in a controlled area, place or room away from others by physically restricting the individual's ability to leave), physical/mechanical restraint (includes the use of mechanical devices such as cuffs, belts, chains or hands-on immobilisation to prevent free movement of a person) (Smithard & Randhawa, 2022) and chemical restraint (use of medications against the person's will to control a person's movement and/or behaviour, not within a standard treatment or dosages used to treatment medical conditions) (Stark, 2018).

Restrictive care practices have been argued by some to be a form of abuse and violation of human rights (An et al., 2016; Steinert & Lepping, 2009) and considered a sign of a mental health crisis arising from a lack of adequate numbers of staff or other conditions, all indicating poor quality of care (Chien et al., 2005; Wright, 2017). As a result, different strategies, guidelines and policy reforms have been introduced to reduce and eliminate (when possible) the use of RCPs (Duxbury et al., 2019; Gerace & Muir-Cochrane, 2019; Kinner et al., 2017) through the advocation of least restrictive, recovery-oriented care (Chisholm et al., 2023; Recupero et al., 2011). On the other hand, restraint and seclusion are claimed to be efficacious in preventing injury and reducing agitation that might cause harm to the patient (McSherry & Maker, 2021) or other people, including staff (Fisher, 1994). However, the case supporting the clinical benefits of using RCPs is weak (Cusack et al., 2018; Sara Ling et al., 2015).

What does this paper contribute to the wider global clinical community?

- The reported proportion estimates of physical restraint, seclusion and chemical restraint in adult inpatient mental health settings were highly variable across studies.
- Differences in the geographical locations of studies conducted, study designs, measurement approaches and definitions of restrictive care practices explained some of the variabilities in the reported proportions of restrictive care practice use.
- Future collaborative works are required to move forward to endorse consensus definitions facilitating consistency in the classification of restrictive care practice techiniques ensuring the validity and reliablilty of measurement and reporting practices to properly understand the use of restrictive care practices in adult inpatient mental health settings.

The physical and psychological harms of restraint and seclusion have been well documented (Whitecross et al., 2013). Falls, bed sores, injuries, immobility, drug reactions (in the case of chemical restraints), diminished psychological well-being, worsening of aggressive behaviour (the commonly reported reason for using RCPs) and other serious medical complications, including death (Mohr et al., 2003) have been reported to be associated with the use of RCPs (Council, 2017; Simpson et al., 2014). Restrictive care practices also adversely affect the healthcare costs, treatment outcomes, functionality and quality of life of patients (McLaughlin et al., 2016; Soininen et al., 2013), which overall impact the personal recovery process (Shumye et al., 2019; Watson et al., 2014). The negative effects of RCP use could extend to the family, caregivers and health professionals who often experience anxiety/psychological

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distress, guilt or self-blame after applying restrictive care to people (Whitecross et al., 2013).

Evidence notes variations regarding how commonly restrictive care practices have been used across regions and between mental health institutions (Chavulak & Petrakis, 2017; Noorthoorn et al., 2016; Thomann et al., 2021). For example, the prevalence of physical restraint among mental health inpatients ranges from 0.3% in the USA (Terrell et al., 2018) to 64.9% in Poland (Lickiewicz et al., 2020). These differences could be due to important differences in policy settings, patient populations characteristics and the trends of clinical practices between these regions (Oostermeijer et al., 2021; Oster et al., 2016). Previous studies have also noted that variations in the scope of definitions of restrictive practices (measurement outcomes), duration and timing of restrictive care practice data collection during the patient admission period (Bowers et al., 2007), measurement methods and inconsistencies in laws, policies and institutional commitments to reduce the use of RCPs explained some of the variations in the proportion estimates reported (Bleijlevens et al., 2016; De Cuyper et al., 2022) across studies. This makes it challenging for researchers and clinicians to make valid comparisons on the rates of restrictive care practice use, and properly evaluate the effectiveness of RCP reduction programmes (Okin, 1985; Thomann et al., 2021). The data inconsistency further creates complications in drawing meaningful conclusions regarding the true nature of RCP use which again impacts the delivery of evidence-based metal health care services (De Cuyper et al., 2022; Paterson & Duxbury, 2007). However, there is an evidence gap in how the reported proportions of people exposed to RCPs are variable across studies and the methodological factors contributing to such variations in inpatient adult mental health settings (Janssen et al., 2013).

1.1 | Aims

This study sought to: (1) synthesise the existing evidence reporting the proportion estimates of people exposed to different forms of restrictive care practices in adult mental health inpatients and (2) investigate potential sources of variability for these proportion estimates in terms of factors related to the population, definitions and data collection approaches of restrictive care practice use.

2 | METHODS

2.1 | Study design

This study was a systematic and meta-analysis of recently published studies, and findings of this review have been reported according to Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) 2020 checklist (Page et al., 2021) (Appendix S1). The PIC (Population, Intervention and Concept/Context) approach was used to address the review concepts, consisting of: (1) population-**P** (psychiatric inpatients/synonymous), (2) intervention-**I** (restrictive care practice/synonymous) and (3) concept/context-**C** (mental health, mental illness and synonymous) (Appendix S2). The protocol for this systematic review was registered to the International Prospective Register of Systematic Reviews-PROSPERO (Ruano et al., 2018): (CRD42022335167).

2.2 | Inclusion criteria

In this review, we used a predefined eligibility assessment criteria, and all studies were evaluated against this criteria. A study was considered eligible if: (1) the data required to generate the proportions of one or more form of RCP (numbers of patients or cases exposed to RCP and total numbers of patient samples) were reported, (2) conducted in adult mental health inpatient settings, or inpatient data was separately reported for studies that included both inpatient and outpatient populations, (3) the research design was cross-sectional survey, cohort/longitudinal, retrospective, randomised controlled trials, pre/post-tests, trials or quasi-experimental study, and (4) the study has been published between 1 January 2010 and 15 August 2022. For chemical restraint, there were several studies where phenomena that could potentially be classified as chemical restraint were referred to using other ambiguous or vague descriptions. As a result, we included studies only if they used the term 'chemical/pharmacological restraint' or clearly stated that the primary reason(s) for the medication administration was to restrain the patient or manage behaviour. In other words, studies were excluded if the primary intention/s of the medication administration was as a treatment modality to cure medical symptoms, or if the primary intention for medication administration was not clearly stated.

Exclusion: A study was excluded if: (1) necessary data to calculate the proportion of at least one of the three review outcomes (physical/mechanical restraint, seclusion and chemical restraint) was not reported, (2) the study was published in non-English languages or the English language translation was available, (3) the study was conducted at forensic, paediatric or geriatric psychiatric populations, nursing home care settings or people with disability or (4) it was a letter, newspaper, magazine article, commentary, legal/ethics paper, systematic review or case study.

2.3 | Searching methods

Search strategies for this systematic review were set through discussion and an iterative review process with pilot database searches to ensure that known relevant papers were being captured through the search strategy. The search was undertaken from six databases (MEDLINE, PsycINFO, CINAHL, Scopus, Embase and Web of Science) using keywords, medical subject headings and MeSH terms in a PIC approach. Studies that have been published in the English language and available online from 1 January 2010 to 15 August 2022 were included. In addition, the reference lists of all included articles were manually checked for the availability of potentially eligible studies to be included in this systematic review and meta-analysis.

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2.4 | Study selection

Once the electronic database searches were completed, all retrieved articles were exported to EndNote (Babineau, 2014) and then imported to the Covidence software (Babineau, 2014) for screening purposes. Initially, duplicate articles were excluded, and title/abstract screening was conducted by one author (ZB) with frequent support and supervision from the other co-authors. Then, two authors (ZB and JC), independently conducted the full-text review using a predefined eligibility assessment criterion developed by all authors. During the full-text review, studies with necessary quantitative data that could be used to calculate the proportions of people exposed to physical/mechnical restraint, seclusion or chemical restraint were considered eligible. Conflicts between the two reviewers' decisions were resolved by involving a third author (TH) using the same assessment criteria.

2.5 | Outcome measurement and data extraction

The outcomes selected for this systematic review were:

1. The proportion of patients reported to have been exposed to physical/mechanical restraint
2. The proportion of patients reported to have been exposed to seclusion
3. The proportion of patients reported to have been exposed to chemical restraint
4. Factors contributing to variations in the reported estimates of physical/mechanical restraint, seclusion and chemical restraint

Some studies measured and reported 'mechanical restraint' and 'physical restraint' separately by referring restraints performed using equipment devices and human force/pressure, respectively. However, majority of the included studies used the term 'physical restraint' to broadly refer to restraint performed both by using equipment devices and human forces. In this study, the term 'physical/mechanical restraint' was used to cover restraints that have been applied either via material devices or hands-on immobilisation techiniques.

Necessary data (numerator and denominator) required to calculate the proportion for each review outcome along with the numerator (where available) were primarily extracted. The numerator data were the number of patients who were exposed to some form of restrictive care practices at the time when they received inpatient mental health services, and the denominator was the total sample of patients who participated in the study. For studies that did not directly report the numbers of patients exposed to restrictive care practices or the total number of samples, these numbers were derived from the reported prevalence or percentages (if available) to enable the pooling of proportions based on the numerators and denominators across all studies included in the meta-analysis. In some instances, authors were contacted to obtain the relevant data to calculate the proportion estimates of people exposed to restrictive care practices.

The author team also extracted data related to factors that might contribute to data variability in the reported proportions of patients exposed to different forms of restrictive care practice between studies. These factors included:

1. Geographical locations of the study conducted (Europe, North America, Asia, Australia, South America and Africa)
2. Study design (retrospective, prospective-longitudinal, cross-sectional and trial)
3. Data collection approaches (extraction of data from electronic medical databases, manual review of patient medical records/notes, reviewing unspecified sources of medical records, interview of patients/staff, direct staff/researcher observation, a combination of more than one method from 1 to 5, unspecified/unclear measurement approaches)
4. Presence or absence of explicit definitions of restrictive care practice techiniques applied in the study
5. Inclusion or exclusion of specific intervention practices/techiniques in the given definitions of restrictive care practices (scope of definitions)
6. Type of restraint techniques/materials included in the given definitions RCPs (devices/equipment, human force/pressure, both devices and human pressure or unclear/unspecified).

Regarding the inclusion and exclusion of specific interventions in the classification of RCPs, some studies used broad definitions that encompass more than one intervention practice (e.g. '…the use of mechanical devices including safety vests, cuffs, straps, pelvic holders, waists, or belts…'), while others had a narrow focus and incorporated only one technique in their applied definitions of RCPs (e.g. physical restraint was performed by manual holding the person's arms…). We made comparisons in the reported proportions of RCP use between these studies based on the differences in the classifications of different intervention practices to be included in the categories of RCP within their applied definitions.

Data extraction was performed by one author (ZB) with frequent review and cross-checking of a sample of 15% of papers with other co-authors (TPH, MP, DCL). Data were extracted into a custom-developed spreadsheet prepared in a Microsoft Excel format. Modifications to the formatting of the spreadsheet took place during the review process to accommodate different data classification options that became necessary as data from more papers were extracted.

2.6 | Risk of bias assessment

The author team examined the risk of bias for all included studies using the Joanna Briggs Institute (JBI) Critical Appraisal Checklist for prevalence studies. All included studies were separately evaluated using nine questions with four rating options ('Yes', 'No',

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'Unclear' or 'Not applicable'). The JBI assesses the possibility and extent of biases related to methodological qualities, including design, sampling procedures, descriptions of study settings, data collection and analysis techniques that have been applied by studies (Santos et al., 2018).

2.7 | Data analysis

The author team reviewed and cleaned the data before being exported to STATA software (version 17.0) for analysis. The standard errors for the proportion of physical/mechanical restraint, seclusion and chemical restraint were independently calculated for each primary study using the binomial distribution formula. Meta-analysis of binomial data was performed to pool the proportion estimates of people exposed to physical/mechanical restraint, seclusion and chemical restraint using data reported across studies. We used a random effect model with 95% confidence intervals (CI). Data used to calculate the proportion were the number of patients exposed to restrictive care practices (numerator) and the total number of patient samples who participated in the study (denominator). For pre and post-test designs, and cohort/longitudinal studies that measured restrictive care practices multiple times, the baseline/initial data was used to estimate the proportion. In the case of trial and experimental studies, the number of patients exposed to restrictive care practices and the number of total study samples in both the control and intervention groups before the intervention was added.

In the meta-analysis, the I^2 statistic was used to express the data heterogeneity (Fletcher, 2007). Heterogeneity in meta-analysis refers to variations, discrepancies or inconsistencies in measurement outcomes between studies (Ioannidis, 2008). The I^2 statistic estimates the proportion of the variance in study estimates that is due to heterogeneity (von Hippel, 2015). Higher values of the I^2 statistic indicate increasing data heterogeneity and vice versa. The author team used a random-effects meta-analysis model in consideration that the restrictive care practice measurement approaches being used in the included studies were part of a broader range of possible ways that these measurements could take place. The inclusion of the additional error term incorporated in calculating confidence intervals for pooled estimates using the random effects model was justified.

Sub-group analyses were undertaken to visualise differences in the pooled proportion estimates of physical/mechanical restraint, seclusion and chemical restraint between subgroups based on the factors of geographical location, study design, measurement approach, presence/absence of explicit definitions, inclusion/exclusion of different restrictive care practice techniques in the given definitions and type restraint techniques considered in the definition of physical/mechanical restraint.

Meta-regression was computed to estimate how much each of the above subgroup factors impacted the estimated proportions of restrictive care practice use. Meta-regression is an appropriate analysis model to investigate whether the effects of explanatory variables had statistically significant impacts on the estimates of the outcome variables or not (Dias et al., 2013). The regression coefficient (B) is used to estimate how a variable in each subgroup affects the outcome estimate as compared to the nominated reference subgroup (Baker et al., 2009). An alpha threshold of $p<.05$ was used to interpret the meta-regression analysis outputs in terms of whether a factor significantly explained variability in the reported measurement outcome data (Thompson & Higgins, 2002).

Meta-regression also allowed the calculation of a residual I^2 value, indicating the extent to which the subgrouping factor explained variability in the measurement outcome data. However, a limitation of the I^2 statistic is that I^2 increases with the number of patients included in the studies in a meta-analysis. The adjusted R^2 and τ^2 statistics were calculated to address this concern. The adjusted R^2 estimates the proportion of variance in the outcome explained by the model while adjusting for the number of variables included in the model. The square root of τ^2 was calculated as its estimates do not systematically increase with either the number variables (Jackson & Bowden, 2016) or the size of studies included in a meta-analysis (Rücker et al., 2008).

3 | RESULTS

3.1 | Search results

A total of 4309 studies were retrieved from the electronic database searches, and 1861 of them were excluded due to duplications. After screening title/abstracts for the remaing 2448 studies, 200 studies were considered for further full-text reading review. During the full-text review, 97 articles were further excluded for various reasons, while 103 studies fulfilled the inclusion criteria. In addition, four studies were identified from the reference lists of included studies, yielded a total of 77 studies to be included in this meta-analysis (Figure 1). Some studies reported data for more one of the review outcomes (physical/mechanical restraint and seclusion physical/mechanical restraint and chemical restraint, seclusion and chemical restraint or all the three outcomes). As a result, the sum of studies included in the meta-analysis of each outcome is greater than 77. The complete reference lists of all studies included in the meta-analysis have been provided as an additional file (Appendix S3).

3.2 | Characteristics of included studies

Physical/mechanical restraint: There were 59 studies [32 from Europe, 14 from Asia, seven from North America, four from Australia, one from South America and another one from Africa] included in the meta-analysis of physical/mechanical restraint. Nearly half ($n=29$) of the studies collected the physical/mechnaical restraint data through the interrogation of electronic databases. Among reviewed studies, 30 provided explicit definitions of physical/mechnical restraint. However, there were variations in the scope

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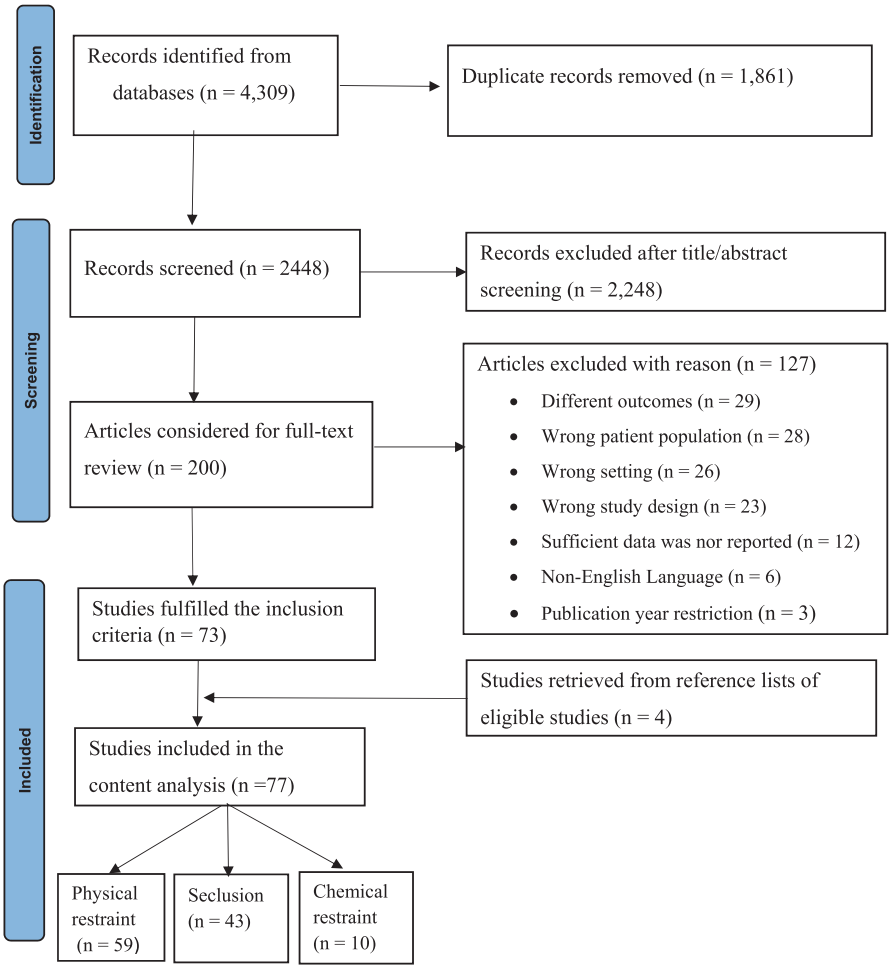

FIGURE 1 Flow chart of literature search using PRISMA. Some studies reported data for more than one form of restrictive care practices and the sum of all articles included the meta-analysis for each outcome is greater than 77. [Colour figure can be viewed at wileyonlinelibrary.com]

of definitions of physical/mechanical restraint to include or exclude specific restraint practices techiniques within the classifications of physical restraint in their applied definitions. For example, 22 studies had broader definitions that included more than one intervention practice to be classified in the family of physical/mechanical restraint, while eight studies used narrower definitions, and considered only one intervention practice to be included within their applied definitions of the measurement outcome (physical/mechnical restraint) (Table 1).

Seclusion: There were 43 studies [20 from Europe, four from Asia, eight from North America, other eight from Australia, two from Africa and one from South America] included in the meta-analysis of seclusion. Nearly half ($n = 21$) of the studies used the interrogation of electronic databases to collect seclusion data from secondary data sources. Among reviewed studies, 31 studies provided explicit definitions for seclusion. The author team noted discrepancies in the descriptions studies used to define seclusion to include or exclude specific seclusion techniques (places, areas or rooms) in the classification seclusion as per their applied definitions. Some definitions ($n = 14$) considered more than one alternative place/area where seclusion can be implemented (e.g. empty room, units in the ward, controlled area or purposefully designed rooms), while the definitions used by four studies considered only one seclusion technique (e.g. confinement of a person in a sparsely furnished room) (Table 2).

Chemical restraint: There were 10 studies [six from Europe, two from Asia and other two from North America] included in the meta-analysis to estimate the pooled proportion of patients exposed to chemical restraint. All studies provided definitions of chemical restraint, and stated that the primary intention of medication administration was to restrain the patient, but not for standard treatment modality. However, concepts incorporated across the given definitions of chemical restraint were highly variable. Some definitions specified medication categories that should be considered to apply chemical restraint. However, the types and number of medication classes included across these definitions were inconsistent (Table 3).

3.3 | Meta-analysis results

3.3.1 | Pooled proportion estimate of people exposed to physical/mechanical restraint

The reported proportion estimates of patients exposed to physical/mechanical restraint were highly variable across included studies,

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TABLE 1 Characteristics of studies included in the meta-analysis to estimate the pooled proportion of people exposed to physical/mechnaical restraint in adult mental health inpatients.

Study/ publication year	Country	Study design	Data collection/ measurement approaches	Presence of explicit definitions* of physical restraint (Y/N)	More than one technique was included in the definition (Y/N)	Type restraint techniques incorporated with the definition of physical restraint	Total sample of patients	Number of patients exposed to physical restraint	Prevalence of physical restraint
Bilanakis et al. (2010)	Greece	Retrospective	Manual review of patient medical notes	Yes	No	Devices	282	5	1.8%
De Hert et al. (2010)	Belgium	Retrospective	Interview Unspecified source of medical records	Yes	Yes	Devices	679	138	20.3%
Diaz et al. (2010)	USA	Retrospective		No	No	Unspecified	11	24	21.6%
Hendryx et al. (2010)	USA	NRCT*	Unspecified source of medical records	Yes	No	Human force	1266	115	9.0%
Husum et al. (2010)	Norway	Retrospective	Direct observation	Yes	Yes	Devices	1214	117	9.6%
Dumais et al. (2011)	Canada	Retrospective	Manual review of patient medical notes	Yes	Yes	Devices	2721	476	17.5%
Di Lorenzo et al. (2012)	Italy	Retrospective	Interrogation of electronic data sources Manual review of patient medical notes	Yes	Yes	Both devices and human force	1610	268	16.6%
Bowers et al. (2012)	UK	Cross-sectional	Manual review of patient medical notes	No	No	Unspecified	522	36	6.9%
Huf et al. (2012)	Brazil	RCT**	Direct observation	Yes	No	Devices	556	51	9%
Tarsitani et al. (2013)	Italy	Longitudinal	Manual review of patient medical notes	No	No	Unspecified	200	14	7%
Nakamura et al. (2013)	Japan	Cross-sectional	Interrogation of electronic data sources	No	No	Devices	5556	1731	31.1%
Noda et al. (2013)	Japan	Retrospective	Interrogation of electronic data sources	Yes	Yes	Both devices and human force	1232	148	12%
Knutzen et al. (2013)	Norway	Retrospective	Interrogation of electronic data sources	Yes	No	Devices	3365	306	9%

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TABLE 1 (Continued)

Study/ publication year	Country	Study design	Data collection/ measurement approaches	Presence of explicit definitions* of physical restraint (Y/N)	More than one technique was included in the definition (Y/N)	Type restraint techniques incorporated with the definition of physical restraint	Total sample of patients	Number of patients exposed to physical restraint	Prevalence of physical restraint
Gonzalez et al. (2014)	Spain	Retrospective	Direct observation	No	No	Unspecified	87	38	43.7%
Zhu et al. (2014)	China	Longitudinal	Interrogation of electronic data sources	Yes	No	Devices	160	82	51.2%
Mah et al. (2015)	Canada	Interventional	Interrogation of electronic data sources	No	No	Human force	115,383	7476	6.5%
Guzman et al. (2015)	Spain	Retrospective	Unspecified sources of medical records	Yes	No	Devices	550	100	18.2%
Hadi et al. (2015)	Iran	Retrospective	Interrogation of electronic data sources	No	No	Unspecified	607	186	30.6%
Wu (2015)	Hong-Kong	Retrospective	Interrogation of electronic data sources	Yes	Yes	Devices	335	133	39.3%
An et al. (2016)	China	Cohort	Interview	Yes	Yes	Devices	1364	371	27.2%
Flammer et al. (2016)	German	Retrospective	Interrogation of electronic data sources	Yes	No	Devices	1194	119	9.9%
Guzman et al. (2016)	Spain	Retrospective	Unspecified sources of medical records	Yes	Yes	Both devices and human force	1428	174	12.2%
Hilger et al. (2016)	German	Retrospective	Unclear	No	No	Devices	6244	469	7.5%
Noorthoorn et al. (2016)	Netherland	Retrospective	Interrogation of electronic data sources	Yes	Yes	Both devices and human force	5169	379	7.3%
Andersen et al. (2017)	Denmark	Prospective	Interrogation of electronic data sources	Yes	Yes	Both devices and human force	224	25	11.2%
Brady et al. (2017)	Australia	Cross-sectional	Interview	No	No	Unspecified	1825	79	4.3%
Jegede et al. (2017)	USA	Retrospective	Interrogation of electronic data sources	Yes	No	Devices	3255	84	2.6%

(Continues)

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TABLE 1 (Continued)

Study/ publication year	Country	Study design	Data collection/ measurement approaches	Presence of explicit definitions* of physical restraint (Y/N)	More than one technique was included in the definition (Y/N)	Type restraint techniques incorporated with the definition of physical restraint	Total sample of patients	Number of patients exposed to physical restraint	Prevalence of physical restraint
McLeod et al. (2017)	Australia	Retrospective	Interrogation of electronic data sources	No	No	Unspecified	232	25	10.7%
Barnett (2018)	Malawi	Retrospective	Manual review of patient medical notes	Yes	Yes	Both device and human force	419	130	31%
Hotzy et al. (2018)	Switzerland	Cohort	Interrogation of electronic data sources	No	No	Devices	612	11	1.8%
Fukasawa et al. (2018)	Japan	Retrospective	Interrogation of electronic data sources	Yes	Yes	Devices	10,013	1496	14.9%
Gowda et al. (2018)	India	Prospective	Interview	Yes	Yes	Devices	467	94	20.1%
Hsu and Chan 2018	Taiwan	Unclear	Interrogation of electronic data sources	No	No	Unspecified	200	59	29.5%
Reitan et al. (2018)	Norway	Retrospective	Manual review of patient medical notes	Yes	Yes	Both devices and human force	19,283	324	1.6%
Terrell et al. (2018)	USA	Retrospective	Manual review of patient medical notes	No	No	Unspecified	2946	9	0.3%
Odgaard et al. (2018)	Denmark	Longitudinal	Interrogation of electronic data sources	Yes	Yes	Devices	218	35	16%
Silić et al. (2018)	Hungary	Retrospective	Unclear	Yes	No	Unspecified	1235	122	9.9%
Saeed et al. (2019)	Pakistan	Retrospective	Unspecified sources of medical records	Yes	No	Devices	477	70	14.7%
Sampogna et al. (2019)	Italy	Follow-up	Manual review of patient medical notes	Yes	Yes	Both devices and human force	294	26	8.8%
Danielsen et al. (2019)	Denmark	Cohort	Interrogation of electronic data sources	Yes	Yes	Devices	5050	100	2%

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TABLE 1 (Continued)

Study/ publication year	Country	Study design	Data collection/ measurement approaches	Presence of explicit definitions* of physical restraint (Y/N)	More than one technique was included in the definition (Y/N)	Type restraint techniques incorporated with the definition of physical restraint	Total sample of patients	Number of patients exposed to physical restraint	Prevalence of physical restraint
Lykke et al. (2019)	Denmark	Retrospective	Interrogation of electronic data sources	Yes	Yes	Devices	1698	35	2.0%
Hamann et al. (2020)	German	RCT**	Interrogation of electronic data sources	No	No	Unspecified	154	35	22.7%
Lickiewicz et al. (2020)	Poland	Cohort	Direct observation	Yes	Yes	Devices	450	173	38.4%
Poloni et al. (2020)	Italy	Retrospective	Interrogation of electronic data sources	Yes	No	Devices	65	35	53.8%
Whitecross et al. (2020)	Australia	Cross-sectional	Interrogation of electronic data sources Manual review of patient medical notes	No	No	Devices	981	61	6.2%
Hirose et al. (2021)	Japan	Case control	Interrogation of electronic data sources	Yes	Yes	Devices	223,285	29,403	13%
Guzman et al. (2021)	Spain	Follow-up	Unspecified sources of medical records	Yes	No	Devices	17,330	2567	14.8%
El-Abidi et al. (2021)	Spain	Cross-sectional	Interrogation of electronic data sources	Yes	No	Devices	464	119	25.6%
Bjerg et al. (2021)	Germany	Prospective	Interview	No	No	Unspecified	411	71	31.6%
Chieze et al. (2021)	Switzerland	Retrospective	Interrogation of electronic data sources	Yes	No	Unspecified	253	7	2.8%
O'Callaghan et al. (2021)	Ireland	Cross-sectional	Interview	Yes	No	Human force	107	10	9.3%
Pérez et al. (2021)	Spain	Retrospective	Interrogation of electronic data sources	Yes	Yes	Devices	3318	412	12.4%
Wang et al. (2021)	China	Cross-sectional	Interrogation of electronic data sources	No	No	Unspecified	981	145	14.8%

(Continues)

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TABLE 1 (Continued)

Study/ publication year	Country	Study design	Data collection/ measurement approaches	Presence of explicit definitions* of physical restraint (Y/N)	More than one technique was included in the definition (Y/N)	Type restraint techniques incorporated with the definition of physical restraint	Total sample of patients	Number of patients exposed to physical restraint	Prevalence of physical restraint
Kuppili et al. (2022)	India	Cross-sectional	Unspecified sources medical records	Yes	No	Devices	339	2	0.6%
Smith et al. (2022)	USA	Retrospective	Interrogation of electronic data sources	No	Yes	Both devices and human force	12,977	961	7.4%
Välimäki et al. (2022)	Hong Kong	Prospective	Interview Unspecified sources of medical records	Yes	No	Devices	4170	1798	43.1%
Ward et al. (2022)	Australia	Retrospective	Interrogation of electronic data sources Interview	No	No	Unspecified	438	77	17.6%

Note: Explicit definitions* = Definitions that describe the conceptual domain or constructs of physical/mechnical restraint using straightforward statements (e.g. physical restraint is defined as...or ... called..... or ..... refers to...).

Abbreviations: NRCT*, non-randomised control trial; RCT**, randomised control trial.

ranging from 0.3% to 54%. The pooled estimate for physical/mechanical restraint was 0.14 [95% CI: (0.13, 0.16)]. The I^2 statistic for this pooled analysis was 99.7%, with a Cochran's Q of $p < .001$, indicating considerable data heterogeneity across the pooled studies (Figure 2).

3.3.2 | Factors that explained data heterogeneity in the pooled proportion of people exposed to physical/mechanical restraint

Based on the results of the meta-regression analysis, geographical locations and data collection/measurement approaches found to have statistically significant contribution to the variations in the reported proportion estimates of people exposed to physical/mechanical restraint. Studies from North America [0.08, 95% CI: (0.05, 0.12)], Australia [0.09, 95% CI: (0.05, 0.14)] and Europe [0.12, 95% CI: (0.10, 0.13)] had significantly lower pooled proportion estimates compared to studies from Asian countries [0.24, 95% CI: (0.19, 0.29)]. Geographical locations explained 17.0% of the variations in the pooled proportion estimates of physical/mechnical restraint ($r^2 = 17.07$).

Studies that collected physical/mechanical restraint data through direct observation [0.24, 95% (0.12, 0.36)] had a significantly higher pooled proportion estimates of physical/mechanical restraint in comparison with those that collected physical/mechanical restraint data using manual review of patient medical notes/records [0.06, 95% CI: (0.04, 0.07)]. However, factors including study design, type of definition, number of restraint techniques included in the applied definition, types of restraint materials included in the definitions of physical/mechanical restraint and presence/absence of explicit definitions of physical/mechanical restraint in the study did not show statistically significant contributions to the variations in this pooled proportion estimate (Table 4).

3.3.3 | Pooled proportion estimate of people exposed to seclusion

The reported proportion estimates of people exposed to seclusion was highly variable across studies included in this review, ranging from 0.2% to 56%. The pooled estimate for seclusion was 0.15 [95% CI: (0.14, 0.16)]. The I^2 statistic for this pooled analysis was 99.5% with Cochran's Q of $p < .001$, indicating considerable data heterogeneity across the pooled studies (Figure 3).

3.3.4 | Factors that explained data heterogeneity in the pooled proportion of people exposed to seclusion

Based on the results of the meta-regression analysis, geographical location, study design and scope of definitions of seclusion explained some of the variabilities in the reported proportion estimates of people exposed to seclusion. Studies from North America

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TABLE 2 Characteristics of studies included in the meta-analysis to estimate the pooled proportion of people exposed to seclusion in adult mental health inpatients.

Study/publication year	Country	Study design	Data collection/ measurement approach	Presence of explicit definition* of seclusion (Y/N)	Inclusion of more than one technique in the definition of seclusion (Y/N)	Total sample of patients	Number patients exposed to seclusion	Prevalence of patients exposed to seclusion
Prinsloo and Noonan (2010)	Ireland	Retrospective	Manual review of patient medical notes Interrogation of electronic data sources	Yes	Yes	339	122	35.9%
Bilanakis et al. (2010)	Greece	Retrospective	Manual review of patient medical notes	Yes	Yes	282	25	8.9%
Raboch et al. (2010)	Europe	Retrospective	Interrogation of electronic data sources	Yes	No	2030	122	6.0%
De Hert et al. (2010)	Belgium	Retrospective	Interrogation of electronic data sources Interview	No	No	679	170	25%
Diaz et al. (2010)	USA	Retrospective	Interrogation of electronic data sources	No	No	111	18	16.21%
Hendryx et al. (2010)	USA	RCT**	Unspecified sources of medical records	Yes	No	1266	28	2.1%
Husum et al. (2010)	Norway	Retrospective	Observation	Yes	Yes	1214	424	34.9%
Lee et al. (2010)	Australia	Cross-sectional	Interrogation of electronic data sources	Yes	No	7338	214	2.9%
Dumais et al. (2011)	Canada	Retrospective	Interrogation of electronic data sources Manual review of patient medical notes	Yes	Yes	2721	632	23.2%
Bowers et al. (2012)	UK	Cross-sectional	Manual review of patient medical notes	No	Yes	522	39	7.5%
Vruwink et al. (2012)	Netherland	Retrospective	Interrogation of electronic data sources	Yes	Yes	385	159	41.3%
Huf et al. (2012)	Brazil	NRCT*	Observation	No	No	556	54	9.5%
Tyrer et al. (2012)	New Zealand	Retrospective	Unclear	No	Yes	254	23	9.0%
Larue et al. (2013)	Canada	Prospective	Interview	Yes	No	251	50	19.9%
Janssen et al. (2013)	Netherland	Retrospective	Interrogation of electronic data sources	Yes	No	5097	718	14%
Griffith et al. (2013)	UK	Retrospective	Interrogation of electronic data sources	Yes	No	347	96	27.7%

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TABLE 2 (Continued)

Study/publication year	Country	Study design	Data collection/ measurement approach	Presence of explicit definition* of seclusion (Y/N)	Inclusion of more than one technique in the definition of seclusion (Y/N)	Total sample of patients	Number patients exposed to seclusion	Prevalence of patients exposed to seclusion
Whitecross et al. (2013)	Australia	Cross-sectional	Interrogation of electronic data sources Interview	No	No	1077	158	14.7%
Nawka et al. (2013)	Europe	Cross-sectional	Interrogation of electronic data sources	Yes	No	522	56	10.5%
Noda et al. (2013)	Japan	Retrospective	Interrogation of electronic data sources	Yes	Yes	1232	687	55.8%
Bullock et al. (2014)	Australia	Retrospective	Interrogation of electronic data sources	No	No	469	88	18.8%
Chiba and Subramaney (2015)	South Africa	Retrospective	Unspecified sources of medical records	Yes	No	612	112	18.3%
Mah et al. (2015)	Canada	RCT**	Interrogation of electronic data sources	No	No	115,383	6544	5.7%
Noorthoorn et al. (2016)	Europe	Retrospective	Interrogation of electronic data sources	Yes	No	306,064	32,512	7.6%
Flammer et al. (2016)	German	Retrospective	Interrogation of electronic data sources	Yes	No	1194	35	2.9%
Brady et al. (2017)	Australia	Cross-sectional	Interview	Yes	No	1825	150	8.2%
Jegede et al. (2017)	USA	Retrospective	Interrogation of electronic data sources	Yes	No	3255	95	2.9%
Chavulak et al. (2017)	Australia	Retrospective	Unspecified sources of medical records	Yes	Yes	655	79	12%
McLeod et al. (2017)	New Zealand	Prospective	Interrogation of electronic data sources	Yes	Yes	7239	782	10.8%
Verlinde et al. (2017)	Netherland	Longitudinal	Interrogation of electronic data sources	Yes	No	2244	352	15.7%
Hotzy et al. (2018)	Switzerland	Cohort	Interrogation of electronic data sources	No	No	612	89	14.5%
Fukasawa et al. (2018)	Japan	Retrospective	Interrogation of electronic data sources	Yes	Yes	10,013	3679	36.7%
Gowda et al. (2018)	India	Prospective	Interview	No	No	467	85	18.2%
Terrell et al. (2018)	USA	Retrospective	Manual review of patient medical notes	No	No	2843	96	3.4%

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TABLE 2 (Continued)

Study/publication year	Country	Study design	Data collection/ measurement approach	Presence of explicit definition* of seclusion (Y/N)	Inclusion of more than one technique in the definition of seclusion (Y/N)	Total sample of patients	Number patients exposed to seclusion	Prevalence of patients exposed to seclusion
Barnett (2018)	Malawi	Retrospective	Manual review of patient medical notes	Yes	Yes	419	127	30.3
Jury et al. (2019)	New Zealand	Retrospective	Interrogation of electronic data sources	Yes	Yes	11,341	842	7.4%
Saeed et al. (2019)	Pakistan	Retrospective	Unspecified sources of medical records	Yes	No	477	20	4.2%
Sampogna et al. (2019)	Italy	Longitudinal	Manual review of patient medical notes	Yes	No	294	20	6.8%
Bjerg et al. (2021)	German	Retrospective	Interview	No		333	60	18%
O'Callaghan et al. (2021)	Ireland	Cross-sectional	Interview	Yes	No	107	9	8.4%
Haefner et al. (2021)	USA	Experimental	Manual review of patient medical notes	Yes	Yes	388	23	5.9%
Chieze et al. (2021)	Geneva	Retrospective	Interrogation of electronic data sources	Yes	No	865	288	32.3%
Vruwink et al. (2022)	Netherland	Retrospective	Interrogation of electronic data sources	Yes	No	1106	184	16.7%
Flammer et al. (2022)	German	Retrospective	Interrogation of electronic data sources	Yes	No	105,782	8796	38.%

Note: Explicit definitions* = Definitions that describe the conceptual domain or constructs of seclusion using straightforward statements (e.g. seclusion is defined as…or …called…… or …… refers to…).
Abbreviations: NRCT*, non randomised control trial; RCT**, randomised control trial.

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TABLE 3 Characteristics of studies included in the meta-analysis to estimate the pooled proportion of people exposed to chemical restraint in adult mental health inpatients.

Study/publication year	Country	Study design	Data collection/measurement approaches	Presence of explicit definition* of chemical restraint (Y/N)	Inclusion of more than one drug category in the definition of chemical restraint (Y/N)	Total sample of patients	Numbers of patients exposed to chemical restraint	Prevalence of people exposed to chemical restraint
Bilanakis et al. (2010)	Greece	Retrospective	Manual review of patient medical notes	Yes	No	314	30	10.5%
Knutzen et al. (2013)	Norway	Retrospective	Interrogation of electronic data sources	Yes	Yes	3365	65	1.9%
Mah et al. (2015)	Canada	RCT**	Interrogation of electronic data sources	Yes	No	115,383	21,422	18.6%
Shahpesandy et al. (2015)	UK	Retrospective	Unspecified sources medical records	Yes	No	74	24	32.4%
Reitan et al. (2018)	Norway	Retrospective	Manual review of patient medical notes	Yes	Yes	316	201	63.0%
Gowda et al. (2018)	India	Prospective	Patient interview	Yes	No	467	271	58%
Hotzy et al. (2018)	Switzerland	Prospective	Interrogation of electronic data sources	No	No	612	89	14.5%
Saeed et al. (2019)	Pakistan	Retrospective	Unspecified sources of medical records	Yes	No	477	72	15%
Smith et al. (2022)	USA	Retrospective	Interrogation of electronic data sources	No	Yes	12,977	2047	15.7%
Kuppili et al. (2022)	India	Cross-sectional	Unspecified sources of medical records	Yes	No	339	76	22.4%

Abbreviation: RCT**, randomised control trial.

Note: Explicit definitions* = Definitions that describe the conceptual domain or constructs of chemical restraint using straightforward statements (e.g. chemical restraint is defined as…or …called…… or …… refers to…).

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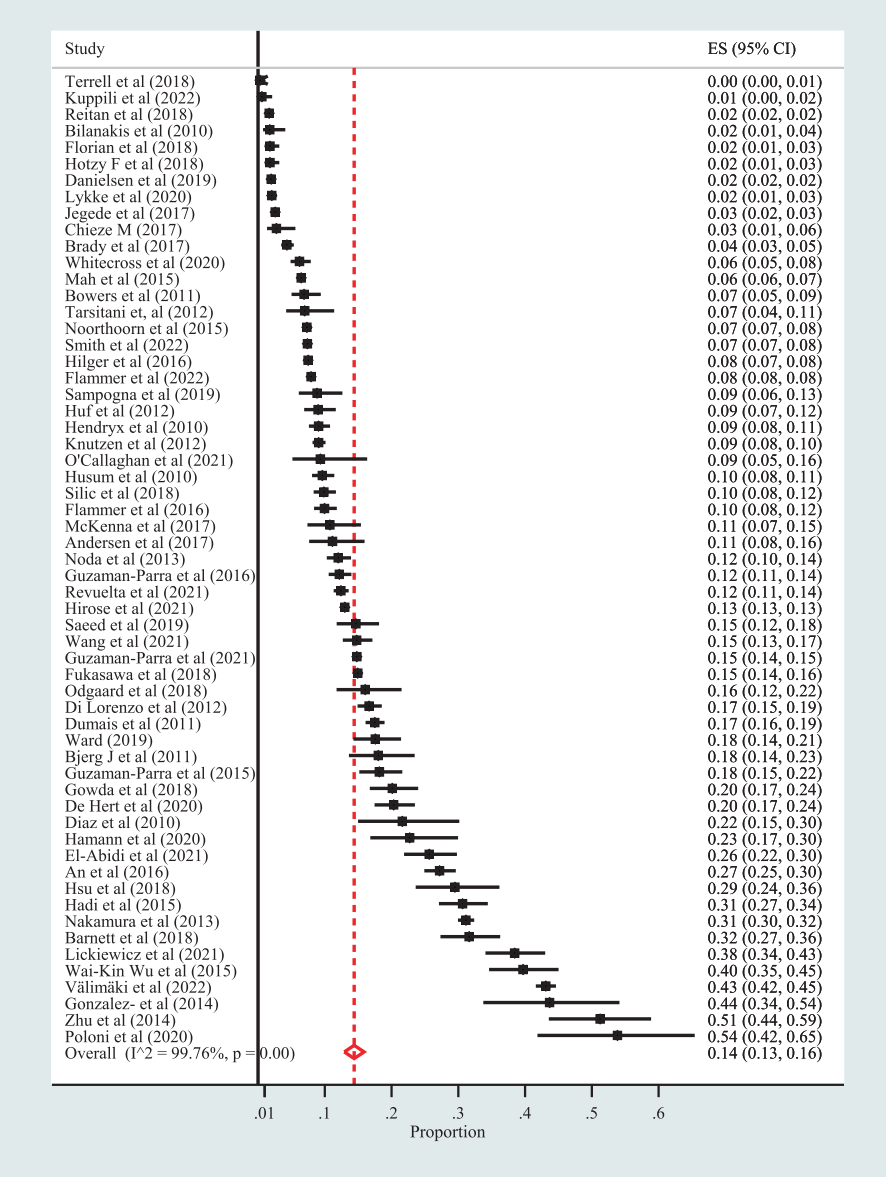

FIGURE 2 Forest plot for the pooled proportion estimate of people exposed to physical/mechnaical restraint in adult mental health inpatients. [Colour figure can be viewed at wileyonlinelibrary.com]

had a significantly lower pooled proportion estimate [0.09, 95% CI: (0.006, 0.12)] compared to studies from Asian countries [0.29, 95% CI: (0.08, 0.49)]. Differences in geographical location of studies contributed to 12.6% variations in the pooled proportion estimate ($r^2 = 12.59\%$). Similarly, trial studies had a significantly lower pooled proportion estimate of people exposed to seclusion [0.06, 95% CI: (0.03, 0.08)] compared to retrospective studies [0.19, 95% CI: (0.17, 0.21)]. Study design explained 9.8% of variations in the reported proportion estimates of seclusion ($r^2 = 9.8\%$). Regarding the inclusion or exclusion of specific intervention practices within the applied definitions of seclusion, studies that included only one technique in the definition of seclusion had a significantly lower pooled proportion estimate of seclusion [0.10, 95% CI: (0.09, 0.11)] compared to studies that included more than one technique of seclusion [0.23, 95% CI: (0.13, 0.30)]. Thus, variation in the scope of definitions of seclusion contributed to the 13.6% of the variability in the pooled proportion estimates of people exposed to seclusion ($r^2 = 13.6\%$). However, data collection methods and the presence/absence of explicit definitions of seclusion did not have statistically significant contributions to the data variability (Table 4).

3.3.5 | Pooled proportion of estimate people exposed to chemical restraint

The reported proportion estimates of people exposed to chemical restraint were highly variable across studies included in this review, ranging from 1% to 58%. The pooled proportion estimate for patients exposed to chemical restraint was 0.25 [95% CI (0.18, 0.31)]. The I^2 statistic for this pooled analysis was 99.8% with Cochran's Q of $p < .001$, indicating considerable data heterogeneity across the pooled studies (Figure 4).

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TABLE 4 Sub-group and Meta-regression analysis results of factors contributing to variations in the proportions of people exposed to physical restraint, seclusion and chemical restraint in adult mental health inpatients

Physical restraint (N=59)			Sub-group analysis results		Meta-regression analysis results				
Variables	Variable categories	No. of included studies in each sub-category	Sub-group pooled proportion [95% CI]	Heterogeneity across studies [I^2]	Coefficient [95% CI]	P-vale	I^2-residual	Tau^2	Adj. R^2
Geographical regions of studies conducted	Asia	14	0.24 [0.19, 0.29]	99.6%	Ref.				
	Europe	32	0.12 [0.10,0.13]	99.4%	-0.112 [-0.187, -0.037]	.04 *	99.5%	0.012	17.0%
	Australia	4	0.09 [0.05, 0.14]	94.7%	-0.145 [-0.277, -0.012]	.032*			
	North America	7	0.08 [0.05, 0.12]	99.8%	-0.151[-0.259, 0.043	.007*			
	Africa	1	0.32 [0.27, 0.36]	---	0.225 [-0.025, 0.476]	.077			
	South America	1	0.09 [0.07, 0.12	---	-0.151 [-0.392, 0.088]	.211			
Study designs	Prospective	15	0.18 [0.14, 0.22]	99.7%	Ref.				
	Retrospective	32	0.14 [0.12, 0.15]	99.5%	-0.030 [-0.110, 0.050]	.455	99.6%	0.015	-2.9%
	Cross-sectional	8	0.12 [0.04, 0.21]	99.6%	-0.057 [-0.169, 0.054]	.309			
	Trial	4	0.10 [0.07, 0.13]	92.0%	-0.064 [-0.209, 0.080]	.375			
Data collection/ measurement approaches	Manual review of patient medical notes	7	0.06 [0.04. 0.07]	98.2%	Ref.				
	Interrogation of electronic databases	29	0.14 [0.12, 0.16]	99.7%	0.076 [-0.030, 0.182]	.135	99.6%	0.014	0.3%
	Interview	5	0.16 [0.04, 0.27]	98.9%	0.076 [-0.071, 0.224]	.303			
	Direct observation	4	0.24 [0.12, 0.36]	98.3%	0.161 [0.001, 0.321]	.048*			
	Unspecified sources of medical records	6	0.12 [0.05, 0.18]	99.4%	0.036 [-0.010, 0.173]	.31			
	Combination of more than one method	6	0.20 [0.08, 0.32]	99.6%	0.120 [-0.019, 0.260]	.09			
	Unclear	2	0.08 [0.07, 0.08	---	0.005 [-0.196, 0.206]	.961			
Presence of explicit definitions* of physical restraint	No	20	0.18 [0.12, 0.24]	95.1%	Ref.				
	Yes	30	0.15 [0.13, 0.17]	99.7%	-0.016 [-0.086, 0.053]	.645	99.7%	0.015	-1.4%
Type of restraint techniques included in the definition of physical restraint	Only equipment devices	30	0.17 [0.14, 0.20]	99.7%	Ref.				
	Only human force	4	0.08 [0.06, 0.09]	99.3%	-0.092 [-0.226, 0.042]	.176	99.5%	0.015	-0.7%
	Both devices and human force	9	0.12 [0.08, 0.15]	99.3%	-0.053 [-0.149, 0.042]	.267			
	Unclear	16	0.14 [0.11, 0.17]	98.5%	-0.026 [-0.105, 0.052]	.498			

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TABLE 4 (Continued)

Physical restraint (N=59)			Sub-group analysis results		Meta-regression analysis results				
Numbers restraint techniques included in the definition physical restraint	More than one	22	0.16 [0.13, 0.19]	99.8%	Ref.				
	Only one	19	0.12 [0.10, 0.14]	99.5%	-0.015 [-.065, 0.096]	.176	99.7%	0.014	-2.4%
	Unspecified	18	0.15 [0.11, 0.19]	99.7%	-0.020 [-0.105, 0.065]	.267			
Seclusion (N=43)									
Geographical regions of studies conducted	North America	8	0.09 [0.06, 0.12]	99.0%	Ref.				
	Europe	20	0.16 [0.14, 0.17]	99.8%	0.073 [-0.023, 0.169]	.133	99.1%	0.012	12.6%
	Asia	4	0.29 [0.08,0.49]	99.7%	0.189 [0.048, 0.330]	.010*			
	Australia	8	0.10 [0.07, 0.13]	98.8%	0.006 [-0.112, 0.125]	.912			
	Africa	2	0.22 [0.20, 0.25]	---	0.144 [-0.038, 0.327]	.119			
	South America	1	0.10 [0.07, 0.12]	---	-0.002 [-0.246,0.241]	.983			
Study designs	Retrospective	26	0.19[0.17, 0.21]	99.6%	Ref.				
	Prospective	7	0.15 [0.12,0.18]]	92.4%	-0.043 [-0.143, 0.057]	.379	99.5%	0.013	9.8%
	Cross-sectional	6	0.09 [0.05, 0.13]	97.5%	-0.105 [-0.210, 0.001]	.051			
	Trial	4	0.06 [0.03, 0.08]	96.2%	-0.133 [-0.250, 0.008]	.037*			
Data collection/ measurement approaches	Interrogation of electronic databases	21	0.16 [0.14, 0.18]	99.7%	Ref.				
	Manual review of patient medical notes	6	0.10 [0.05, 0.15]	96.9%	-0.063 [-0.179, 0.052]	.274	99.5%	0.015	-3.5%
	Interview	5	0.14 [0.09, 0.20]	93.2%	-0.023 [-0.148, 0.102]	.710			
	Combination of more than one method	5	0.22 [0.16, 0.28]	96.8%	0.015 [-0.072, 0.177]	.40			
	Direct observation	2	0.21[0.19, 0.23]	---	0.053 [-0.131, 0.239]	.561			
	Unspecified sources of medical records	3	0.08 [0.01, 0.15]	---	-0.086 [-0.240, 0.067]	.263			
	Unclear	1	0.09 [0.06, 0.13]	---	-0.077 [-0.335, 0.179]	.543			
Presence of explicit definitions* of seclusion	Yes	31	0.16 [0.15, 0.18]	99.6%	Ref.				
	No	12	0.12 [0.10, 0.16]	96.5%	-0.059 [-0.156, 0.038]	.228	99.5%	0.014	-0.8%
Number of seclusion techniques included in the definition	More than one	14	0.23 [0.13, 0.30]	99.7%	Ref.				
	Only one	4	0.10 [0.09,0.11]	99.5%	-0.117 [-0.195, -0.038]	.005*	99.4%	0.012	13.6%
	Unspecified	21	0.12 [0.08, 0.17]	96.9%	-0.097 [-0.227, 0.032]	.136			
	No definition	4	0.19 [0.16, 0.45]	89.1%	-0.042 [-0.173, 0.088]	.512			

(Continues)

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TABLE 4 (Continued)

Physical restraint (N=59)			Sub-group analysis results		Meta-regression analysis results				
Chemical restraint (N=10)									
Geographical regions of studies conducted	North America	2	0.18 [0.18,0.18]	---	Ref.				
	Europe	6	0.24 [0.08, 0.40]	99.4%	0.056 [-0.348, 0.461]	.751	99.2%	0.043	-7.3%
	Asia	2	0.32 [0.06, 0.57]	---	0.230 [-0.265, 0.727]	.309			
Study designs	Prospective	2	0.27 [0.24, 0.29]	---	Ref.				
	Retrospective studies	6	0.23 [0.14, 0.32]	99.7%	0.157 [-0.677, 0.363]	.498	99.6%	0.048	-17.9%
	Others	2	0.19 [0.18,0.19]	---	-0.157 [-0.557, 0.293]	.489			
Data collection/ measurement approaches	Manual review of patient medical notes	2	0.26 [0.23, 0.29]	---	Ref.				
	Interrogation of electronic databases	6	0.22 [0.14, 0.30]	99.8%	-0.151 [-0.574, 0.271]	.426	99.8%	0.047	-16.5%
	Unspecified sources of medical records	2	0.17 [.0.13, 0.20]	---	-0.134 [-0.655, 0.387]	.562			
Presence of explicit definitions* of chemical restraint	Yes	8	0.27 [0.19, 0.36]	99.8%	Ref.				
	NO	2	0.16 [0.15, 0.16]	---	-0.125 [-0.503, 0.253]	.467	99.8%	0.042	-5.1%
Number of drug categories included in the definition of chemical restraint	Only one	7	0.21 [0.12. 0.30]	99.8%	Ref.				
	More than one	3	0.34 [0.09, 0.59]	---	0.123 [-0.204, 0.451]	.410	99.8%	0.041	-2.8%

Note: Explicit definitions* = Definitions that describe the conceptual domain or constructs of restrictive car practices using straightforward statements (e.g..... is defined as...or ...called..... or ..... refers to.....). * = Statistically significant ($p < .05$) at a 95% confidence interval, showing variables that explained some of the significant data hetrigeniety in the reported proportion of restricitve care practice use.
Abbreviations: Adj. R^2, adjusted R-squared; CI, confidence interval; I^2, percentage of data heterogeneity; Ref, refers to reference variables used to compare the effects of other variables on the reported data variability.

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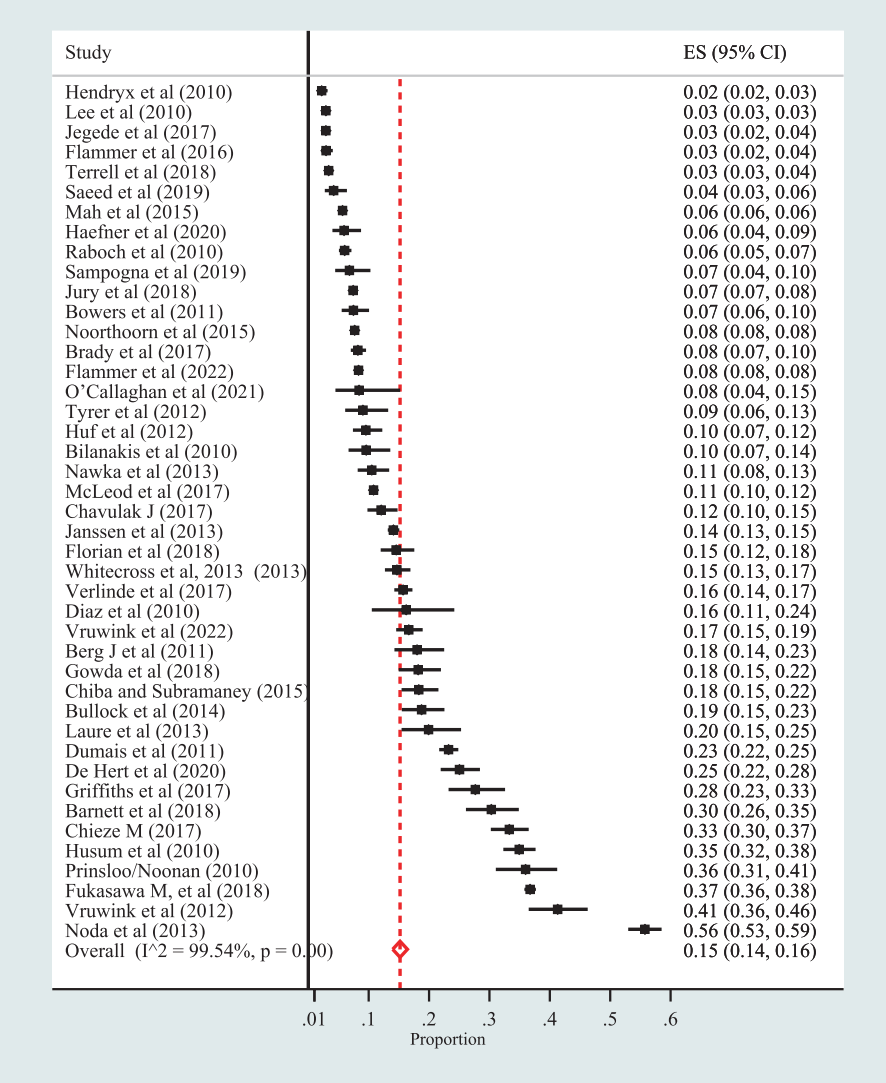

FIGURE 3 Forest plot for the pooled proportion estimate of people exposed to seclusion in adult mental health inpatients. [Colour figure can be viewed at wileyonlinelibrary.com]

3.3.6 | Factors that explained data heterogeneity in the pooled proportion of people exposed to chemical restraint

The pooled proportion estimates of people exposed to chemical restraint were clustered by geographic location, study design, data collection/measurement methods, presence/absence of explicit definitions of chemical restraint and inclusion/exclusion of specific drug categories within the given definitions of chemical restraint. However, none of these factors had statistically significant contributions to the variations in the proportion estimates of people exposed to chemical restraint (Table 4).

4 | DISCUSSION

This systematic review and meta-analysis identified that the pooled proportions of adult mental health inpatients exposed to physical/mechanical restraint, seclusion and chemical restraint were 14.4%, 15.8% and 25.6%, respectively. These data were drawn from studies published since 2010, indicating that through a range of policy reforms, recommendations (De Cuyper et al., 2023) and guidelines that have been published before or since this time advocate for the minimisation of restrictive care practice use (Degener & Koster-Dreese, 1995; WHO, 2019). It is possible that these policies and strategies may have had the desired effect (Muir-Cochrane & Holmes, 2001; Recupero et al., 2011) on the reported proportion estimates of restrictive care practice use. However, these estimates represent the 'floor' that restrictive care practice minimisation strategies are working toward (Duxbury et al., 2019).

Some studies reported the use of multiple forms of restrictive care practices on one person at a time. There were three pair-wise combinations possible (e.g. chemical + physical/mechanical, physical/mechanical + seclusion, seclusion+ chemical) and an overall combination of all three (La Vonne et al., 2007). The author team have choosen not to analyse or present findings using these combinations for the sake of clarity of messages and that there were few papers that reported each specific combination. The analysis focused on the proportion of patients exposed to the three different forms of restrictive care practices separately (physical/mechanical

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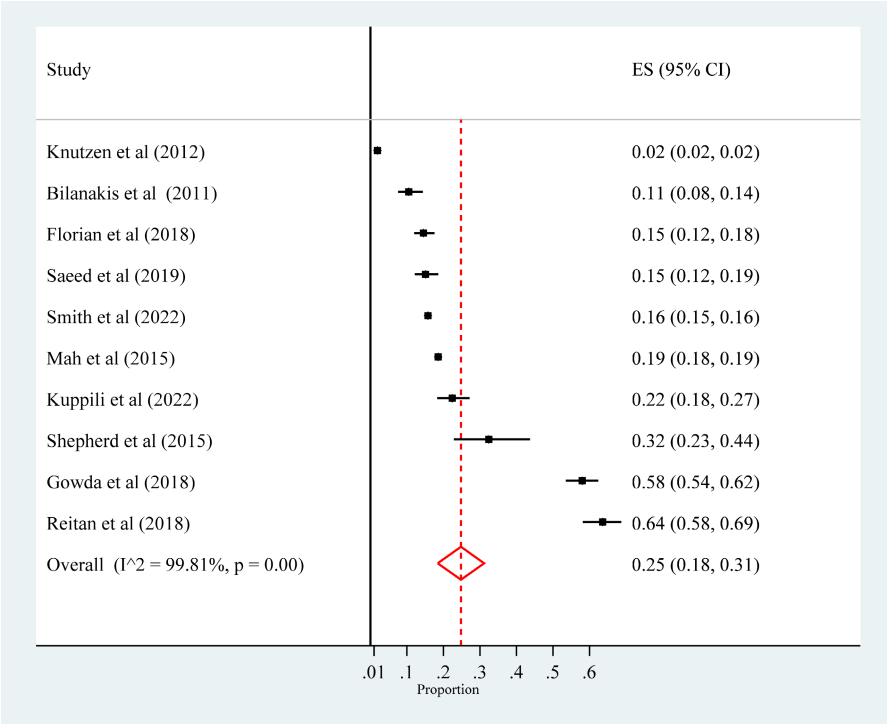

FIGURE 4 Forest plot for the pooled proportion estimate of people exposed to chemical restraint in adult mental health inpatients. [Colour figure can be viewed at wileyonlinelibrary.com]

restraint, seclusion and chemical restraint). Therefore, an overall proportion of people exposed to any form of restrictive care practice would be the arithmetic sum of the three pooled estimates as some individual patients could be concurrently exposed to multiple categories of restrictive care practices.

A key problem identified in this review in trying to understand whether policies, recommendations, guidelines or other reduction strategies have the desired effects to minimise restrictive care practice use is the variability in approaches used to define, measure and report these practices. Such inconsistencies in the measurement procedures of restrictive care practice impact the reported estimates of restrictive care practice use, raising concerns regarding the reliability and validity of the data (Haines et al., 2009). For example, studies that used direct observation to collect physical/mechanical restraint data reported an absolute difference (0.16 higher than those that used manual review of patient medical records) in the proportion of physical restraint. This makes direct comparison across this literature to draw inferences about the success of different policy settings or approaches more complex (Savage et al., 2024), and creates considerable uncertainty in the pooled estimates of the proportion of patients exposed to different forms of restrictive care practices generated in the current meta-analysis (Ahmed & Ishtiaq, 2021).

The data heterogeneity in results reported between studies included in this review was explained to a greater or lesser extent by the variables investigated. The largest drop in I^2 was observed in the analysis when investigating the geographical location for both physical/mechnaical restraint and seclusion data ($\tau^2 = 0.012$). The I^2 drop indicates the extent to which data heterogeneity/variability in the reported measurement outcomes is explained by independent variables (factors). That means a variable with a higher drop in I^2 during meta-regression has a higher contribution to the variability in the reported proportion estimate (Higgins & Li, 2022). The second τ^2 drop for physical/mechanical restraint was recorded when investigating approaches used to measure physical/mechanical restraint data ($\tau^2 = 0.014$). For seclusion data, the second largest drop in τ^2 in the current analysis was observed when investigating the study design ($\tau^2 = 0.013$). For chemical restraint data, the largest τ^2 drop was occurred when investigating the number of drug categories included within the applied definitions of chemical restraint ($\tau^2 = 0.041$). The presence or absence of explicit definitions of chemical restraint showed the second largest τ^2 drop ($\tau^2 = 0.042$) for chemical restraint data. These discrepancies indicate that the tools used to understand factors that explain the variability in reported measurement outcomes of studies included in a meta-analysis have inconsistencies that are potentially attributable to their individual limitations. There were several studies with very large sample sizes in the current meta-analysis, which in this context, makes it less likely to find the I^2 results as trustworthy for this purpose. In this review, study design, data collection approaches and scope of definitions of measurement outcomes (inclusion or exclusion of different forms of restrictive care techniques) are factors that researchers have control over and appear to have considerable influence on reported results.

Regarding geographical locations, studies from Asian countries had the highest proportion estimates (0.24 for physical/mechanical restraint and 0.29 for seclusion). Differences in geographical locations of studies contributed to 17.1% and 12.6% of the variations in the proportion estimates of people exposed to physical/mechanical restraint and seclusion, respectively. Compared with studies from Asian countries, the proportion of physical/mechanical

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restraint decreased by 16%, 15% and 12% in studies from North America, Australia and Europe, respectively. Such variations could be explained by the differences in the level of innovative policies and strategies aimed to minimise restrictive care practice use (Hook & Bogdanov, 2021; Maramis et al., 2011), or the critically low numbers of mental health professionals and a relatively unsophesticated mental health care systems in most Asian countries compared to North America, Australia and Europe (Sharan et al., 2017). For example, fewer psychiatrists per-population exist in most Asian countries (Yates, 2018). Individual and institutional variations in the preferences of the methods used to measure, document and/or report restrictive care practice incidents across different geographical regions (Pérez-Revuelta et al., 2021) or healthcare services could also explain some of these variations (El-Abidi et al., 2021). This illustrates a need to understand the changing nature of mental health services and adapt similar documentation (Barnett, 2018) and reporting cultures to enhance ongoing monitoring and evaluation of restrictive care practice reduction strategies (McKeown et al., 2020).

In this review, the author team also noted discrepancies across studies in the applied definitions of physical/mechnaical restraint, seclusion and chemical restraint. For example, some studies used broader definitions of seclusion that encompass various techniques of seclusion, quiet room, units in the ward, controlled area or purposefully designed places, while others used narrower definitions and measured only a single intervention practice which could be a sub-set of the brader definitions. Such inconsistencies in the definitions and classifications of restrictive care practice techniques make it difficult to make valid comparisons between these studies as they have measured different number and type of restricitve care practice techiniques, but reported their findings using a similar umbrellla term (Roper et al., 2015). This suggests the need for future research works to maintain consistencies in the definition or classification, measurement and reporting cultures of restrictive care practice episodes across different clinical settings or jurisdictions and between individuals (Bowers et al., 2007; Robins et al., 2021).

Overall, this comprehensive systematic review identified the integrations/interrelation of different concepts, including study designs, definitions, measurement/data collection methods and reporting protocols to contribute to the variations in the data reported associated with proportions of people exposed to different forms of restrictive care practice use in adult mental health inpatients. For example, the choice of study design can influence the scope and depth of the data being collected, which in turn, can impact the reported proportion estimates the measurement outcome (Mekuriaw et al., 2023). Furthermore, the review highlights the significance of definitions and classification taxonomies in shaping the data reported regarding the proportion estimates of RCP use. The existence of broader or narrower definitions of restrictive care practice techniques can introduce variations in what is being measured and reported, which might lead to inconsistencies in the measurement outcome data related to restrictive care practice use (Velentgas et al., 2013). The nature of the intervention practices under investigation could also influence the choice of measurement/reporting methods, yielding inconsistent results and impacting the reported proportions of restrictive care practices (Savage et al., 2024). This overall shows the integration/interelation of inconsistencies in the definition/classification, data collection and reporting methods to affect the validity and reliabilty of RCP data reported across studies.

4.1 | Future implications

This review revealed data inconsistencies in the reported proportion estimates of RCP use stemmed from variations in measuring and defining different forms of RCP techniques. This could be an indication for the inadequacy of the current measurement methods and conceptualisation of RCPs which can introduce potential biases in the reported measurement outcome data. This hinders accurate comparisons for evaluating RCP reduction policies and practices, underscoring the need for an internationally consistent definitions of restrictive practices. Endorsing standardised definitions and uniform reporting protocols of restrictive care practice episodes is crucial in ensuring that data is collected and reported consistently and comparably (Savage et al., 2024). By understanding the interrelationships of definition and reporting practices to affect the measurement outcome data for RCPs, researchers and policymakers can work toward more accurate and standardised measurement and reporting frameworks to enhance the reliability and validity of their research findings (Chavulak et al., 2023). This would facilitate making valid comparisons between studies to inform policymakers, professionals and other stakeholders to plan and deliver evidence-based metal health care (Lepping et al., 2016). Valid comparisons in the use of restrictive care practices based on different policy settings and approaches used internationally can be made only once different regions and researchers use consistent definitions and measurment approaches and mental health workers have common understanding of RCPs (Lay et al., 2011). However, even with this, there is a need for further work to understand if health professionals, researchers and service users have the same understanding of how to classify and report restrictive care practice scenarios, even when provided with standardised definitions. Previous work in fall prevention has identified that health service personnel classified different scenarios as falls at different rates depending on differences in their professional disciplines as well as institutional and workplace-related factors (Haines et al., 2009). It is reasonable to expect that similar variations and discrepancies in how definitions of restrictive care practices are interpreted and applied may also exist in adult mental health inpatient settings.

5 | LIMITATIONS

A key limitation of this study was that we could not pool the proportion estimates of patients exposed to different forms of

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restrictive care practices that were free from statistical heterogeneity. This means that these estimates should be considered in the context of the limited approaches used to generate data across the included studies, and not be used as absolute values for the purposes of clinical benchmarking. Second, the exclusion of studies published in non-English languages, or studies that did not have clear definitions of chemical restraint might create biases, though it is not clear whether they would have had a systematic effect on raising or lowering the pooled estimates. The third limitation might be due to the inclusion of very few studies from Africa and South America. This potentially underpowered contrasts involving in these regions such that a significant difference between subgroups was not identified, even though these regions may have quite different outcomes. It was noted that one study from Africa included in the pooled proportion estimate of people exposed to physical/mechanical restraint data and reported a higher proportion (0.32) than the overall pooled estimate of 0.14. Fourth, the proportion estimates of restrictive care practice use have been calculated by dividing the number of people exposed to restrictive care practices to the total number of samples that participated in the study, but studies that did not report such data were excluded. This means other prevalence calculation methods using occupied bed days and length of hospital stay of the patient could not be applied (Bowers, 2000).

Finally, data regarding some other possible reasons (factors) that might contribute to the discrepancies in the reported proportion estimates of restrictive care practice use (such as participants' diagnosis/disease, their age, health conditions and other socio-demographic factors) have not been consistently reported across included studies. As a result, these factors, as well as other institutional-related factors and policy variations that might contribute to the data hetrogienity in the reported proportion estimates of people exposed to different forms of restrictive care practice techniques could not be investigated.

6 | CONCLUSION

The reported proportion estimates of restrictive care practice use in adult mental health inpatients were highly variable across studies. There appear to be differences between geographical regions in the use of restrictive practices; however, this is complicated by how these prevalence data have been measured and defined. Consistency in defining and measuring restrictive care practices in mental health is required, so that comparisons of the proportion of people exposed to different forms of restrictive care practices between geographical regions, policy settings and practice innovations can be validly made. Once the nature and scale of the experience is more consistently measured and understood, there will be far greater likelihood of establishing and maintaining standards of humane and compassionate care delivery to support and treat people in vulnerable situations when they are most unwell.

7 | RELEVANCE FOR CLINICAL PRACTICES

The reported proportion estimates of people exposed to restrictive care practices in adult inpatient mental health settings were highly variable. Differences in the definition, measurement and reporting cultures of restrictive care practice techniques contributed to these variabilities. This informs researchers, clinicians and healthcare policymakers to move forward to develop consistent definition, documentation and reporting protocols to deliver standardised, least-restrictive and evidence-based mental health care. There is needed to design training programmes and staff support systems, as well as advocacy for policy changes for methodological consistency in the documentation and reporting culture of restrictive care practices. Such changes can have the potential to reduce ethical and legal dilemmas for mental health workers that can occur in part because of difficulties in making decisions in their actual workplaces on which practice should be documented and reported within the classification of restrictive care practices.

AUTHOR CONTRIBUTIONS

TPH, DCL, MP and ZB conceived the idea and developed the review protocol. ZB did database searches with advisory inputs from TPH, DCL and MP. ZB and JC screened the retrieved papers using a predefined eligibility assessment criteria. TH involved in resolving conflicts between the two reviewers' decisions. ZB conducted data extraction with frequent check-ups and support from TPH, DCL and MP. TPH, DCL and ZB conducted statistical data analysis. ZB also wrote the first draft of the manuscript and made several revisions, where MP, TPH and DCL reviewed these subsequent drafts. ZB was responsible in corresponding with the publication process of this paper.

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CONFLICT OF INTEREST STATEMENT

All authors declare that there are no competing interests related to this research work.

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DATA AVAILABILITY STATEMENT

All the data supporting the findings of this review have been reported within the manuscript, and submitted in the supportive files of the manuscript.

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